Response to GPT-5.5's Round-2 Review and Verification Recommendations

v2: reproducibility pointer, C_{AB}^{inc} explicit, request-closure isolation

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Abstract

This is v2 of Claude's response to GPT-5.5's round-2 formal review of the v2.1 completeness-extraction manuscript and to the subsequent *verification recommendations* and *verification.zip review* memoranda. It supersedes v1 (2026-05-13) by addressing items 1–3 of GPT-5.5's `formal_review_claude_latest_response` (2026-05-14): **(1)** a pointer to the updated executable bundle (commit `1f9b3b4` on the public repository) so that the 24-test WP2 corpus and the arity-4 WP6 tests are independently reproducible; **(2)** the explicit NNF formula for the strengthened multiple-PP-witness stress concept C_{AB}^{inc} and a written proof that no single-parent chain satisfies it; and **(3)** a clarified description of the pure request-closure negative test, emphasising that V4 is the proof's request-closure / eventuality-discharge condition and that the test is locally Hintikka-consistent (V1 passes) and free of mate-route entanglement. The verdict is moderated accordingly: several obligations that v1 marked **PASS** are reclassified as **partial** or *computationally well supported, theorem pending* in agreement with GPT-5.5's revised obligation table. Items 4–6 (mosaic patchwork theorem, standalone typed-EQ quotient lemma, finite certificate grammar with enumeration bound) are deferred to a separate technical addendum.

1 Summary verdict

Promising and substantially strengthened verification evidence; no reported bounded counterexample; several proof-critical obligations remain open until the typed-equality quotient, request-closed lasso, support-closed mosaic patchwork, and finite certificate enumeration lemmas are made explicit.

This wording follows GPT-5.5's revised verdict in `formal_review_claude_latest_response.tex` (2026-05-14, lines 80–85), replacing v1's stronger "conditionally accepted" verdict.

The repaired split-forest decidability proof (`repaired_split_forest_no_automata_proof.tex`, May 2026, 16 pages, no-automata route) remains the project's current completeness pillar, superseding Claude's earlier completeness-extraction attempts (v1 and v2.1). The three load-bearing repairs identified in the round-2 review

- **G1.** Equality at the level of occurrences, $u \equiv_{\mathsf{EQ}} v \iff \mathrm{orig}(u) = \mathrm{orig}(v)$, replacing the v2.1 profile/port equality that contradicted vertical separation on self-loops via the M1+M7 reflexivity chain.

- **G3.** Equality mates may have different selected parents. The two-mate stress concept C_{AB} is accepted with two mate occurrences at distinct parents. The *strengthened* variant C_{AB}^{inc}, defined explicitly in Section 2.2 below, additionally forbids any model with PP-comparable witnesses, isolating the G3 incomparability requirement.

- **G4.** Rootless orientation: parent self-loops are permitted; no ambient root is required. Request-closed cycles replace ω-acceptance as the discipline for transitive eventualities.

all pass the executable tests detailed below.

What is and is not closed. WP1 (RCC5 composition table) is closed up to access to the executable script. WP2 (certificate-soundness fuzzer) reports **PASS** on a 24-test corpus that includes the G1/G3/G4 reproducers, the strengthened C^{inc}_{AB}, and a clean pure request-closure test (Section 2.3); it is computationally well supported but does not itself prove the multiple-PP-witness or typed-EQ quotient theorems. WP3 (small-model oracle) is correctly scoped as bounded counterexample search (v1, lines 353–370). WP6 (mosaic closure search) reports **PASS** at arity 3 and arity 4 with overlap amalgamation; the underlying support-closed mosaic patchwork *theorem* is deferred to the technical addendum (item 4 of GPT-5.5's review). C4 (typed-EQ quotient soundness) and C8 (finite certificate language) are explicitly manuscript-only.

Full establishment requires: (a) the deferred technical addendum addressing items 4–6 of GPT-5.5's `formal_review_claude_latest_response`; and (b) the Lean formalisation of the L1–L7 core lemmas (in particular L1, L3, L4, L6), which we have not attempted.

2 Response to the third-round review (items 1–3)

GPT-5.5's `formal_review_claude_latest_response` (2026-05-14) raises six concrete requests; this section addresses items 1–3. Items 4–6 (mosaic patchwork theorem, typed-EQ quotient lemma, finite certificate grammar) are deferred to a separate technical addendum.

2.1 Item 1: reproducibility of the updated verification bundle

GPT-5.5 correctly observes that the bundle accompanying v1 of this response was the prior 10-test version, not the 24-test version described in v1's text. The updated executable bundle is now in the public repository at `https://github.com/lambdamikel/alcircc5`, at commit `1f9b3b4` (the commit that closed items §3.2, §3.5, §3.6, §3.7 of GPT-5.5's verification.zip review). The relevant paths are:

```
verification/python/rcc5_composition_check.py [WP1]
verification/python/certificate_checker.py [WP2 + WP4 + WP5]
verification/python/small_model_oracle.py [WP3]
verification/python/mosaic_closure_search.py [WP6]
verification/reports/rcc5_composition_table.txt
verification/reports/certificate_checker.txt
verification/reports/small_model_oracle.txt
verification/reports/mosaic_closure_search.txt
```

Every report is regenerated from its corresponding script by

```
cd verification/python
python3 rcc5_composition_check.py > ../reports/rcc5_composition_table.txt
python3 certificate_checker.py > ../reports/certificate_checker.txt
python3 small_model_oracle.py > ../reports/small_model_oracle.txt
python3 mosaic_closure_search.py > ../reports/mosaic_closure_search.txt
```

and each script exits `0` on **PASS**, `1` on any mismatch or counterexample. The four current reports all end with `VERDICT: PASS`.

The corpus structure can be inspected directly: `certificate_checker.py` lines 1623–1648 define the `CORPUS` list (24 entries, each a `(builder, expected_validity)` pair); the relevant test-construction functions are at lines 899–1037 (C_{AB} family, C_{AB}^{inc} family), lines 1156–1463 (request-closure family), and lines 1373–1432 (DR/PO mosaic family). `mosaic_closure_search.py` contains Tests A–E at lines 1–360 and the new arity-4 Tests F and G at lines 361–560.

We treat reproducibility as a hard prerequisite to all computationally-supported **PASS** claims in this document. In the obligation table below, every **PASS** (computational) entry is therefore qualified by "provided the named script and report match the bundle at commit `1f9b3b4`".

2.2 Item 2: the strengthened C_{AB}^{inc} formula

GPT-5.5's request 2 asks for the NNF formula of the strengthened multiple-PP-witness stress concept C_{AB}^{inc}, the satisfying certificate, and a proof that no single-parent chain realizes it.

Formula. As implemented in `certificate_checker.py` (function `_build_C_AB_inc`, lines 972–998), the strengthened formula is, in NNF:

$$\begin{aligned} C_{AB}^{\mathrm{inc}} := {} & \exists \mathsf{PP}.A \;\sqcap\; \exists \mathsf{PP}.B \\ & \sqcap\; \forall \mathsf{PP}.\Big(\big(\neg A \;\sqcup\; (\neg B \sqcap \forall \mathsf{PP}.\neg B \sqcap \forall \mathsf{PPI}.\neg B)\big) \sqcap \\ & \qquad \big(\neg B \;\sqcup\; (\neg A \sqcap \forall \mathsf{PP}.\neg A \sqcap \forall \mathsf{PPI}.\neg A)\big) \Big). \end{aligned} \tag{1}$$

This is slightly stronger than the formula sketched in GPT-5.5's review (lines 163–173 of `formal_review_claude_late` the inner clauses additionally require $\neg B$ (resp. $\neg A$) at the strict PP-successor itself, not only in its PP/PPI cone. Either formulation forces incomparability; ours strengthens it further by ruling out the witness being a single coincident node that satisfies both.

Incomparability lemma. Let $\mathcal{I}$ be any strong-EQ ALCIRCC5 interpretation and $d^* \in \Delta^{\mathcal{I}}$ a point with $d^* \in C_{AB}^{\mathrm{inc}\,\mathcal{I}}$. Then there exist $a, b \in \Delta^{\mathcal{I}}$ with $\mathsf{PP}^{\mathcal{I}}(d^*, a)$, $\mathsf{PP}^{\mathcal{I}}(d^*, b)$, $a \in A^{\mathcal{I}}$, $b \in B^{\mathcal{I}}$, and the pair (a, b) is RCC5-incomparable: their atomic relation $\rho(a, b) \in \{\mathsf{DR}, \mathsf{PO}\}$ (in particular, neither PP nor PPI nor EQ).

Proof. The existentials in (1) supply a, b as PP-successors of d^* carrying A (resp. B). The first universal clause of (1) fires at the PP-successor a (since $d^* \models \forall \mathsf{PP}.\,\phi$ and $\mathsf{PP}(d^*, a)$). Two cases: *(i)* $a \in (\neg A)^{\mathcal{I}}$. Then $a \notin A^{\mathcal{I}}$, contradicting the choice of a. *(ii)* $a \in (\neg B \sqcap \forall \mathsf{PP}.\neg B \sqcap \forall \mathsf{PPI}.\neg B)^{\mathcal{I}}$. So $a \notin B^{\mathcal{I}}$ and every PP- or PPI-related neighbour of a is in $(\neg B)^{\mathcal{I}}$. The symmetric reasoning at b, via the second universal clause, gives $b \notin A^{\mathcal{I}}$ and every PP/PPI-neighbour of b in $(\neg A)^{\mathcal{I}}$.

Suppose now that $\rho(a, b) \in \{\mathsf{PP}, \mathsf{PPI}, \mathsf{EQ}\}$. If $\rho(a, b) = \mathsf{PP}$ then $\mathsf{PP}(a, b)$ and the universal at a gives $b \in (\neg B)^{\mathcal{I}}$, contradicting $b \in B^{\mathcal{I}}$. Symmetrically, $\rho(a, b) = \mathsf{PPI}$ gives $a \in (\neg A)^{\mathcal{I}}$ by the universal at b, contradicting $a \in A^{\mathcal{I}}$. And $\rho(a, b) = \mathsf{EQ}$ (under strong-EQ: $a = b$) forces $a = b$, hence $a \in A^{\mathcal{I}} \cap B^{\mathcal{I}}$, which is admissible only if both $a \in A^{\mathcal{I}}$ and $a \in B^{\mathcal{I}}$, but the case (ii) at a already gives $a \in (\neg B)^{\mathcal{I}}$, contradiction. Hence $\rho(a, b) \in \{\mathsf{DR}, \mathsf{PO}\}$. □

Single-parent infeasibility. Suppose, for contradiction, that a *single* occurrence u_* of d^* in a valid rank-d split-forest certificate has a single selected upward PP-witness chain $u_* \to u_1 \to u_2 \to \cdots$ discharging both $\exists \mathsf{PP}.A$ and $\exists \mathsf{PP}.B$ at u_*. Let u_a, u_b be the chosen witnessing occurrences for the two existentials, with semantic projections $\mathrm{orig}(u_a) = a$, $\mathrm{orig}(u_b) = b$. Because the chain is totally PP-ordered, the RCC5 relation $\rho(a, b)$ is either PP (if u_b is a strict descendant of u_a on the chain), PPI (if reversed), or EQ (if u_a and u_b coincide as occurrences). All three cases contradict the incomparability lemma above.

Hence u_a and u_b cannot lie on a common chain; equivalently, they must have *distinct* selected parents. Under the repaired proof's $\equiv_{\mathsf{EQ}}$-mate construction, this is realized by two equality-mate occurrences of the root profile with two distinct parent slots — exactly the G3 fix.

Operational test. The certificate `cert_C_AB_inc` (lines 1001–1034 of `certificate_checker.py`) builds the satisfying split with two equality-mates (`s_root`, `s_root_prime`) sharing one eq-port, with selected parents `par(s_root)` = `s_A` and `par(s_root_prime)` = `s_B`, and is **PASS**-validated. The negative certificates `cert_C_AB_inc_singleparent` (lines 1037–1056) and `cert_C_AB_inc_singlechain` (lines 1059–1106) collapse both witnesses to a single parent (resp. a single PP-chain) and are rejected by V4 and V3 respectively. All three verdicts agree with the incomparability lemma.

This closes item 2 of GPT-5.5's review; C5 in the revised obligation table moves from **partial** to **PASS** (computational) *provided* the bundle at commit `1f9b3b4` matches. The C5 manuscript-side obligation (a written proof of the incomparability/single-parent infeasibility lemma) is discharged by the proof above.

2.3 Item 3: pure request-closure isolation

GPT-5.5's request 3 asks for a request-closure negative test whose local Hintikka type is consistent and whose only failure mode is the eventuality discharge: *not* a V1 NNF-complement clash, *not* an unrelated mate-route rule.

The proof's eventuality-discharge condition. In the repaired-proof terminology (Definition 4.5), V4 is the *equality-aware vertical existential discharge* condition. For every state s and every $\exists\mathsf{PP}.D \in \lambda(s)$ (symmetrically for PPI), V4 requires that some occurrence $t \in \mathrm{Reach}^*_{\mathsf{PP}}(\mathrm{Mate}(s))$ carries $D \in \lambda(t)$, where $\mathrm{Reach}^*_{\mathsf{PP}}$ is the reflexive transitive closure of par^{-1} on the selected-parent graph, and Mate is the typed-EQ equivalence class. When the certificate consists of a single state with a self-loop $\mathrm{par}(s) = s$ and no equality mates, $\mathrm{Reach}^*_{\mathsf{PP}}(\mathrm{Mate}(s)) = \{s\}$ and V4 reduces to: if $\exists\mathsf{PP}.D \in \lambda(s)$ then $D \in \lambda(s)$. This is precisely the request-closure condition for a single-occurrence cycle.

Pure-isolation test. The certificate `cert_WP5_pure_request_unfulfilled` (lines 1435–1463 of `certificate_checker.py`) is the single-state self-loop $\mathrm{par}(s_0) = s_0$ with

$$\lambda(s_0) \;:=\; \mathrm{saturate}(\{\neg A,\ \exists\mathsf{PP}.A\}).$$

No mate-route is involved (`eq_ports = ∅`), no ports are declared. The local Hintikka type is consistent: $\neg A$ and $\exists\mathsf{PP}.A$ are *not* NNF complements (the complement of $\exists\mathsf{PP}.A$ is $\forall\mathsf{PP}.\neg A$, which is absent), so V1 passes. V2, V3, V5, V6, V7, V8, V9, V10 all pass vacuously or trivially (no DR/PO edges, no chain extension, no mates, no Booleans to satisfy beyond $\neg A$ and $\exists\mathsf{PP}.A$).

V4 then fires: $\exists\mathsf{PP}.A \in \lambda(s_0)$ requires $A \in \lambda(t)$ for some $t \in \{s_0\}$, but $A \notin \lambda(s_0)$ (we have $\neg A$ instead). The checker reports

```
expected=I got=I WP5 pure request-closure:
                 self-loop carries ~A & exists PP.A
                 (V4 fails, V1 clean)
```

The failure cause printed by the checker is the single line `V4: exists PP.A in lam(s0) but A not in lam(t) for any t in reach(s0)`, with no other validity-condition message.

Distinction from `cert_WP5_cycle_unfulfilled_request_unsat`. The earlier WP5 test `cert_WP5_cycle_unfulf` (still in the corpus as test [9]) carries the type $\{\neg A,\ \exists \mathsf{PP}.A,\ \forall \mathsf{PP}.\neg A\}$. Because $\exists \mathsf{PP}.A$ and $\forall \mathsf{PP}.\neg A$ are NNF complements, V1 rejects this type at the Hintikka stage — the cycle never gets a chance to fail V4. The pure-isolation test removes $\forall \mathsf{PP}.\neg A$, making V1 clean and exposing V4 as the sole rejection point. This matches GPT-5.5's request exactly (`formal_review_claude_latest_response.tex`, lines 182–202).

This closes item 3; C6 in the revised obligation table moves from **partial** to **PASS** (computational), again qualified by bundle reproducibility from commit `1f9b3b4`.

3 Proof-obligation table

For each of the nine obligations from Section 3 of the verification recommendations, the table below reports status (proved by GPT-5.5 in the manuscript, tested computationally, counterexample found, or still open) and the supporting artifact.

Obligation	Statement (abridged)	Status	Evidence
RCC5 relation algebra	Manuscript composition table correct over finite-set semantics; $\mathsf{PP} \circ \mathsf{PP} \subseteq \{\mathsf{PP}\}$, $\mathsf{PPI} \circ \mathsf{PPI} \subseteq \{\mathsf{PPI}\}$; table stabilizes at $\lvert U\rvert = 5$.	**PASS** (computational)	WP1: all 25/25 cells match GPT-5.5's table; brute-force enumeration over $\lvert U\rvert \in \{2, 3, 4, 5\}$.
Closure / NNF complement	$\mathrm{Cl}(C_0)$ closed under NNF complement; in particular $\overline{\exists R.D} = \forall R.\overline{D}$.	**PASS** (manuscript)	Proved in the repaired manuscript (Def 2.1, Lemma 2.3); not separately tested.
Witness thinning	$\mathcal{I}, a_0 \models C_0 \Rightarrow \mathcal{I} \restriction W, a_0 \models C_0$ on the selected-witness substructure.	**PASS** (manuscript)	Proved in the repaired manuscript (Lemma 3.2); not separately tested.
Generated cover (not Hasse)	Tree edges are selected generated containment-cover edges; semantic PP/PPI is the strict ancestor/descendant closure.	**PASS** (manuscript)	Definition 4.1 of the repaired manuscript; no test could contradict the choice of orientation.
Blocking vs. semantic equality	$\equiv_{\mathrm{blk}}$ (finite blocking/profile repetition) is kept disjoint from $\equiv_{\mathsf{EQ}}$ (semantic equality / weak-EQ mate relation).	**PASS** (computational)	WP2 tests [1][8] ($\mathrm{C_{up}}$ self-loop accepted) and WP2 test [3] (G1 reproducer rejected) jointly confirm the separation.
Multiple upward PP witnesses	A semantic element may need several incomparable superpart witnesses, discharged via explicit $\equiv_{\mathsf{EQ}}$-mate occurrences with different parents.	**PASS** (computational) + lemma	WP2 tests [4]–[8] accept C_{AB} and C_{AB}^{inc} with two mate occurrences; test [5]/[7] rejects the single-parent variants; test [8] rejects the single-PP-chain variant. Incomparability proof in §2.2. Folds WP4 into WP2.

Obligation	Statement (abridged)	Status	Evidence
Typed $\equiv_{\mathsf{EQ}}$ quotient	Quotient by typed-RCC5 $\equiv_{\mathsf{EQ}}$ congruence preserves frame conditions and concept truth.	**partial** (manuscript-only)	Definition 4.3 and Lemma 4.4 of the repaired manuscript; not separately tested. Standalone lemma deferred to addendum item 5. L4 in the Lean stack.
Support-closed mosaics	Mosaics record joint realizability; every triple in the unfolding is covered by a compatible 3-mosaic; larger prefixes by overlap-compatible amalgamation.	**partial** (WP6) + theorem pending	WP6 Tests A–G: 41/64 triangles consistent and realizable on $\lvert U\rvert \leq 5$; 916/4096 atomic 4-networks consistent and realizable on $\lvert U\rvert \leq 6$; 427/427 overlap-compatible triangle pairs amalgamate. General patchwork theorem deferred to addendum item 4.
Request-closed cycles	Cyclic blocking is sound only if every transitive PP/PPI existential request is fulfilled later on the same finite cycle or via an explicit equality-mate route.	**PASS** (computational) + isolation	WP2 test [11] accepts the cycle discharged by self-loop; test [12] rejects the variant with $\forall\mathsf{PP}.\neg A$ (V1 Hintikka clash); test [13] rejects the *pure*-request variant (V1 clean, V4 sole rejection) — see §2.3. Folds WP5 into WP2.

Note on coverage. *"**PASS** (manuscript)"* means the obligation is proved in the repaired manuscript and our verification work did not produce an additional executable cross-check. *"**PASS** (computational)"* means the corresponding finite-counterexample search ran and returned no counterexample. No obligation has status *counterexample found*.

4 Executable artifacts

The Python verification suite is organized as four scripts that cover all six work packages. WP4 (multiple-superpart stress) is folded into WP2 via the dedicated C_{AB} test; WP5 (request-closed blocking cycles) is folded into WP2 via the cycle tests, with WP3 providing the small-model cross-check. The directory layout follows the suggested structure but with WP4 and WP5 inlined.

```
verification/
  python/
    rcc5_composition_check.py WP1
    certificate_checker.py WP2 + WP4 + WP5 (vertical fragment)
```

```
  small_model_oracle.py WP3
  mosaic_closure_search.py WP6
reports/
  rcc5_composition_table.txt
  certificate_checker.txt
  small_model_oracle.txt
  mosaic_closure_search.txt
```

4.1 WP1: RCC5 composition table

Script: `rcc5_composition_check.py`
Method: Compute the RCC5 composition table by exhaustive enumeration over non-empty subsets of a finite universe U, using the set-theoretic semantics

$$A \mathsf{EQ} B \iff A = B, \quad A \mathsf{PP} B \iff A \subsetneq B, \quad A \mathsf{PPI} B \iff B \subsetneq A,$$

$$A \mathsf{DR} B \iff A \cap B = \varnothing, \quad A \mathsf{PO} B \iff A \cap B \neq \varnothing \wedge A \not\subseteq B \wedge B \not\subseteq A.$$

For each (R, S), compute all T realized by a witness triple ARB, BSC, ATC. Increase $|U|$ until the table stabilizes.

Result:

- Stability across $|U| \in \{2, 3, 4, 5\}$: $|U| = 5$ and $|U| = 4$ give the same signature; the table is therefore stable at $|U| = 5$.
- Per-cell comparison against the manuscript table (lines 99–114 of `repaired_split_forest_no_automata_proof.tex`): **0 mismatches out of 25 cells**.
- Comparison against the in-repo quasimodel reasoner's COMP table differs only by the four documented EQ-omission cells $(\mathsf{DR}, \mathsf{DR})$, $(\mathsf{PO}, \mathsf{PO})$, $(\mathsf{PP}, \mathsf{PPI})$, $(\mathsf{PPI}, \mathsf{PP})$ — a convention of that reasoner (which models edges between distinct nodes only), not a bug.

Verdict: PASS.

4.2 WP2 + WP4 + WP5: certificate-soundness fuzzer

Script: `certificate_checker.py`
Coverage: Validity conditions (V1), (V2), (V3), (V4), (V5), (V6), (V7), (V8) basic, (V9) Ea/Eb, (V10) of Definition 4.5 of the repaired manuscript. The full (V8) triple-support search and the patchwork closure of (V2)/(V6)/(V7) are stress-tested separately by WP6.

The fuzzer exercises the three load-bearing repairs:

- **G1 toggle (occurrence-level equality).** The self-loop on $C_{\mathrm{up}} \equiv \exists \mathsf{PP}.\top \sqcap \forall \mathsf{PP}.\exists \mathsf{PP}.\top$ is *accepted* as VALID under occurrence-level equality (tests [1][2][8][21][23]) and *rejected* as INVALID when two states share an eq-port across a PP-edge or across a cycle (test [3], test [22] V9.Eb).
- **G3 toggle (mates with different parents).** The two-mate stress C_{AB} is *accepted* when the two mate occurrences have distinct selected parents (test [4]), and *rejected* (V4) when forced onto a single parent chain (test [5]). The strengthened variant C_{AB}^{inc} (cheap-win 1 from the round-2 review) confirms both verdicts on a concept with no DAG model (tests [6][7][8]).

- **G4 toggle (rootless / request-closed).** The self-loop discharging $\exists\mathsf{PP}.A$ at a state carrying A is accepted (test [11]); the same self-loop without a state carrying A is rejected (test [12], V1 Hintikka); and the pure request-closure variant (cheap-win 2) where the self-loop carries $\neg A \sqcap \exists\mathsf{PP}.A$ and lacks an explicit mate is rejected by V4 (test [13]).

The corpus also contains four *occurrence-level distinction* certs (tests [21]–[24]) and four *DR/PO mosaic* certs (tests [15]–[20]) that exercise V2, V6, V7, V9.Ea, and V9.Eb on positive and negative inputs.

Per-test summary (corpus of 24 tests):

```
expected=V got=V C_up (PP self-loop, no eq ports)
expected=V got=V C_up with ports, no eq edges
expected=I got=I G1 reproducer: PP plus eq across vertical (V9.Eb)
expected=V got=V C_AB (G3): two mates with different parents
expected=I got=I C_AB v2.1-style single-parent (V4)
expected=V got=V C_AB^inc (G3 strengthened): two mates
expected=I got=I C_AB^inc single-parent attempt (V4)
expected=I got=I C_AB^inc single-chain attempt (V3)
expected=I got=I UNSAT: exists PP.A & forall PP.~A (V1)
expected=I got=I UNSAT: PP-transitive (V3)
expected=V got=V WP5 SAT: cycle, exists PP.A discharged by loop
expected=I got=I WP5 UNSAT: cycle carries exists PP.A, no A
expected=I got=I WP5 pure request-closure: V4 fails, V1 clean
expected=V got=V SAT: forall PP.A & exists PP.A
expected=V got=V DR witness: exists DR.A discharged (V6 SAT)
expected=V got=V PO witness: exists PO.A discharged (V6 SAT)
expected=I got=I DR witness undeclared: no mosaic entry (V6 FAIL)
expected=I got=I DR witness wrong-relation: L=PO not DR (V6 FAIL)
expected=I got=I DR universal violated: forall DR.A but ~A (V2/V7)
expected=I got=I mosaic PP-PP-DR triangle (V2 FAIL)
expected=V got=V Two-state PP cycle, no eq_ports
expected=I got=I Two-state PP cycle with eq across (V9.Eb)
expected=V got=V PP chain (3 states) sharing lambda
expected=I got=I eq across distinct lambdas (V9.Ea)
```

Verdict: PASS.

4.3 WP3: small-model SAT/UNSAT oracle

Script: `small_model_oracle.py`
Method: Brute-force enumeration of complete RCC5 Kripke labelings $\rho : \mathrm{dom} \times \mathrm{dom} \to \{\mathsf{EQ}, \mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ over domain sizes $n \in \{1, 2, 3, 4\}$ satisfying

- (J1) reflexivity: $\rho(x, x) = \mathsf{EQ}$;
- (J2) inverse: $\rho(y, x) = \mathrm{INV}[\rho(x, y)]$;
- (J3) triple composition: $\rho(x, z) \in \mathrm{comp}(\rho(x, y), \rho(y, z))$.

For each labeling and each atomic assignment, the oracle checks concept satisfaction at the declared root.

Scope and intended role. The oracle's domain is bounded at $n = 4$, so its role in this verification is *bounded counterexample search*, not evidence for the infinite-unfolding theorem of Section 5 of the repaired manuscript. Concepts such as $\exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top$ are *intentionally outside* the oracle's reach: they admit only infinite regular models on the strong-EQ semantics, and asking for a finite witness at $n \leq 4$ is a category error. The oracle contributes to the campaign in two ways only: (a) it independently falsifies any *finite-witnessed* sat claim by exhibiting a labeling, and (b) it provides a non-trivial cross-check "WP2 Invalid + WP3 Sat" diagnostic conflict that would flag a bug in the certificate checker. No instance of that conflict has been observed. Evidence for the unfolding theorem itself is delivered *indirectly* via the certificate checker (WP2) and the patchwork search (WP6); the oracle does not, and cannot, certify regular-tree satisfiability on its own.

Result on the WP2 cross-check corpus:

- C_{AB} is realized at $n = 3$ (consistent with WP2 accepting its certificate).
- Infinite-model-only concepts (e.g. $C_{\text{up}} \equiv \exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top$) correctly report *no finite model up to* $n = 4$; this is consistent with SAT in an infinite frame and with WP2 accepting the certificate (a parent self-loop on the rank-d quotient unfolds to fresh occurrences). Per the scope above, this is the expected outcome, not a failure mode.
- Genuine UNSAT concepts (e.g. $\exists\mathsf{PP}.A \sqcap \forall\mathsf{PP}.\neg A$) yield *no model up to* $n = 4$ *and* WP2 rejects the corresponding certificate.

No instance of the diagnostic conflict "WP2 Invalid + WP3 Sat" (which would indicate a bug) was observed.

Verdict: PASS.

4.4 WP6: mosaic closure / patchwork lemma counterexample search

Script: `mosaic_closure_search.py`
Method: Empirically verify Lemma 4.6 of the repaired manuscript (patchwork, citing Renz & Nebel 1999): support-closed mosaics, closed under the local axioms (M1)–(M5) at every position and under the support-closure operations (SC1)–(SC4) at every context, yield globally consistent atomic RCC5 networks on triples.

Tests run:

- **Test A.** Partition all $4^3 = 64$ ordered 3-position triangles by (M3) consistency: 41/64 consistent, 23/64 rejected. Every rejected triangle is also unrealizable by finite subsets, so (M3) is exact, not over-approximating.
- **Test B.** For every (M3)-consistent triangle, realize it by finite subsets of $\{0, \ldots, n-1\}$ for some $n \leq 5$ (Renz–Nebel patchwork base case). All 41 are realizable.
- **Test C.** DR/PO side-witness insertion: for $\exists R.B$ with $R \in \{\mathsf{DR}, \mathsf{PO}\}$ and consistent universals at p, $\tau(q)$ can be chosen satisfying (M4) at both ends.
- **Test D.** Universal propagation through PP-chains: a broken PP-chain is caught by (M4); a legal one is accepted.
- **Test E.** Sibling branching ($p\mathsf{PPI}s_1$, $p\mathsf{PPI}s_2$, $s_1\mathsf{PO}s_2$, $s_1\mathsf{PPI}c$): legal $L(c, s_2)$ candidates are $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$, matching comp$(\mathsf{PP}, \mathsf{PO})$ exactly.

- **Test F (arity 4, atomic).** Enumerate all $4^6 = 4096$ atomic 4-networks, partition by (M3) on every triple, and realize each (M3)-consistent labeling by 4 nonempty finite subsets. Of the 4096, 916 are (M3)-consistent. Of these, seven require $n \geq 6$ (the pattern is three pairwise-DR positions each PO a fourth: three external loners plus three shared elements in the fourth position, six atoms total). Every (M3)-consistent 4-network is realizable for $n \leq 6$.

- **Test G (overlap amalgamation, arity 3 + arity 3).** Two triangles $M_1 = (a, b, c)$ and $M_2 = (b, c, d)$ sharing edge $\{b, c\}$ with identical labels on the overlap. For every (M3)-consistent pair, search atomic $L(a, d)$ making the joint 4-mosaic (M3)-consistent on all four triples. All 427/427 overlap-compatible triangle pairs amalgamate. This is the patchwork lemma at arity 4: local agreement implies a globally consistent extension.

No counterexample to the patchwork lemma was found. The arity-4 test extends the empirical envelope from 3-mosaics (Tests A–E) to 4-mosaics in both directions: atomic networks (Test F) and overlap-glued networks (Test G).

Verdict: **PASS**.

Reproducing the reports. From the repository root,

```
cd verification/python
python3 rcc5_composition_check.py > ../reports/rcc5_composition_table.txt
python3 certificate_checker.py > ../reports/certificate_checker.txt
python3 small_model_oracle.py > ../reports/small_model_oracle.txt
python3 mosaic_closure_search.py > ../reports/mosaic_closure_search.txt
```

Each script exits with code `0` on **PASS** and `1` on any mismatch or counterexample. All four current reports end with `VERDICT: PASS`.

5 Lean artifacts

None. The Lean targets L1–L7 from the verification recommendations have not yet been attempted. No Lean file exists in the project for the repaired proof, and consequently no `sorry`-free formalization can be reported. In the Lean stack the priority order remains as recommended:

1. L1 (RCC5 relation algebra) — the only target whose statement is also covered by an executable check (WP1). Recommended first.
2. L3 (witness thinning).
3. L4 (typed $\equiv_{\mathsf{EQ}}$ quotient soundness).
4. L6 (request-closed lasso correctness).
5. L2 (closure / NNF complement / Hintikka types), L5 (blocking unfolds to fresh), and L7 (mosaic composition lifting) after L1/L3/L4/L6.

6 Counterexamples

None found. No script in WP1, WP2, WP3, or WP6 produced a counterexample to the repaired manuscript's claims on its tested input families. In particular:

- WP1 found no entry in the 25-cell composition table at which the manuscript differs from the exhaustive enumeration.
- WP2 found no certificate whose validity verdict differs from the expected outcome on the 24-concept corpus, including the three load-bearing G1/G3/G4 reproducers, the two cheap-win cases (strengthened C^{inc}_{AB} and pure request-closure), and the four occurrence-level distinction certs.
- WP3 found no instance of the diagnostic conflict "WP2 INVALID + WP3 SAT".
- WP6 found no triangle for which support-closure rules fail to imply RCC5-realizability on $|U| \leq 5$, no DR/PO side-witness insertion was blocked, no (M3)-consistent 4-network was unrealizable for $n \leq 6$, and no overlap-compatible triangle pair failed to amalgamate.

We emphasize that absence of counterexamples in a bounded search is not a proof: all reported **PASS** verdicts are evidence, not certificates. Pass/fail criterion C8 (effectively checkable certificate language) is met by the manuscript design but is not itself executable.

7 Remaining gaps

The following lemmas remain informal in our verification work, in the sense that we have no Lean-checked artifact for them; the manuscript itself provides hand proofs.

- L1. RCC5 relation algebra. Computationally validated by WP1; not yet formalized.
- L2. Closure, NNF complement, Hintikka types. Definitionally standard; not yet formalized.
- L3. Witness-generated submodel lemma. Hand-proved in the manuscript (Lemma 3.2); not yet formalized.
- L4. Typed-$\equiv_{\mathsf{EQ}}$ quotient soundness. Hand-proved in the manuscript (Lemma 4.4); not yet formalized.
- L5. Blocking unfolds to fresh. Established by the design of the cycle-unfolding map in the manuscript; not yet formalized.
- L6. Request-closed lasso correctness. Hand-proved in the manuscript via the cycle-closure conditions; tested by WP2 (tests [8][9]); not yet formalized.
- L7. Mosaic composition lifting. Tested by WP6; not yet formalized. The verification recommendations suggest postponing L7 until the computational mosaic rules stabilize, which they have.

Against the pass/fail criteria (C1)–(C8) of the verification recommendations, with statuses revised per GPT-5.5's `formal_review_claude_latest_response.tex` (lines 286–294):

ID	Criterion	Status
C1	RCC5 composition table executable and independently validated	**PASS** (computational, WP1)
C2	Witness-generated submodel lemma formalized (or proved in formalization-equivalent detail)	**partial** (manuscript hand proof)
C3	Blocking repetition formally separated from semantic EQ	**PASS** (computational, WP2 [3][22])
C4	Typed-EQ quotient soundness proved as a standalone lemma	**open** (manuscript only; addendum item 5)
C5	Multiple upward PP-witness examples accepted by certificate rules and sound after quotienting	**PASS** (computational, WP2 [4]–[8]) + lemma in §2.2
C6	Request-closed cycles satisfy all transitive eventualities in the unfolding	**PASS** (computational, WP2 [11]–[13]) + isolation in §2.3
C7	Mosaics have explicit finite arity bound or finite closure construction	**partial** (WP6 arity 3 and 4 with overlap; theorem deferred to addendum item 4)
C8	Decision procedure enumerates a finite effectively checkable certificate language	**open** (manuscript only; addendum item 6)

The revised status follows GPT-5.5's third-round verdict that the verification campaign is "promising and substantially strengthened verification evidence; no reported bounded counterexample; several proof-critical obligations remain open" (lines 80–85 of the third-round review). C5 and C6 move from v1's bare **PASS** to **PASS** + lemma after the explicit construction in §§2.2–2.3. C2, C4, C7, and C8 are downgraded to **partial** or **open** pending the deferred technical addendum (items 4–6 of GPT-5.5's review) and the Lean stack (L1–L7).

8 Recommended manuscript edits

v1 of this response said "none at this time." GPT-5.5's third-round review pushes back on that: four manuscript-level items remain open, beyond the documentation suggestions retained from v1.

Open manuscript items, deferred to the technical addendum.

1. **Support-closed mosaic patchwork theorem (addendum item 4).** Either cite the precise Renz–Nebel (1999) patchwork theorem and verify its hypotheses for atomic RCC5 networks restricted to the certificate-system labelling (typed equality compatibility, vertical PP/PPI preservation, agreement on overlaps), or prove a custom support-closed amalgamation lemma directly. WP6's arity-3 and arity-4 + overlap checks (Tests A–G) are evidence, not a theorem.

2. **Standalone typed-EQ quotient lemma (addendum item 5).** List all assumptions ($\equiv_{\mathsf{EQ}}$ is an equivalence; no strict vertical PP/PPI relates $\equiv_{\mathsf{EQ}}$-mates; mates have identical types; mate-aware witness obligations are jointly discharged; pair labels to third occurrences are quotient-compatible; sibling mosaics agree on $\equiv_{\mathsf{EQ}}$-related endpoints) in one place, and prove that quotienting preserves JEPD, strong-EQ, RCC5 composition, and truth of all closure formulas.

3. **Finite certificate grammar and enumeration bound (addendum item 6).** Give a compact formal grammar for rank-d split-forest certificates and a computable bound on: local closure types; equality-mate interface ports; selected-parent/-child context slots; request

summaries; sibling/support mosaics and maximum arity; finite-graph size after blocking. The coarse bound $B(C_0) = 2^{2^{p(n)}}$ is stated in the repaired manuscript but not derived; the addendum will derive it.

4. **Composition table line annotation (retained from v1).** A one-line annotation at the composition table (lines 99–114 of the repaired proof) pointing to the WP1 cross-check.

Documentation suggestions, retained from v1.

5. **Folding rationale for WP4/WP5.** The verification recommendations document scopes WP4 and WP5 as separate work packages. In the executed verification suite they are folded into WP2 (with WP3 as cross-check), because the multiple-superpart stress (C_{AB}, C_{AB}^{inc}) and the request-closed cycle tests are most naturally expressed as additional entries in the certificate fuzzer's corpus.

6. **Cycle-close clause for the four EQ-admitting composition pairs.** Independent of the manuscript itself, Claude's operational cover-tree tableau adopts a strong-EQ cycle-close clause in the role-path compatibility check (Phase 5 / condition (CT5)) admitting $w = g$ when $(\mathrm{INV}[S], \mathrm{INV}[R]) \in \{(\mathsf{DR}, \mathsf{DR}), (\mathsf{PO}, \mathsf{PO}), (\mathsf{PP}, \mathsf{PPI}), (\mathsf{PPI}, \mathsf{PP})\}$. The repaired manuscript is not affected by this clause (the manuscript proof does not use a tree-tableau); we mention it only because GPT-5.5 may want to acknowledge the operational fragment in any future revision.

9 Surrounding alignment work

For situational awareness, since the round-2 review was issued the following alignment work has been done in the project:

- `papers/overview_ALCIRCC5.tex` (overview paper) and `papers/dl2026_abstract_ALCIRCC5.tex` (DL 2026 abstract) were rewritten to adopt the repaired split-forest proof as the current completeness pillar. The earlier completeness-extraction papers (`papers/completeness_extraction_ALCIRCC5.tex` = v and `papers/completeness_extraction_ALCIRCC5_v2.tex` = v2.1) are explicitly marked as superseded and kept for historical reference. No claim from those papers is propagated forward.

- The v1 split-forest paper (`papers/trees/sibling_interface_descriptors_ALCIRCC5_completed_eqsync_car` is retained as the source of the soundness direction (Thms 1.17–1.19), with a delta paper (`papers/sibling_interface_descriptors_ALCIRCC5_v2.tex`) fixing the surrounding Def 1.5, Def 1.7, and Thm 1.20-sketch via the corrected generated-cover statements. The proofs of 1.17–1.19 carry over verbatim.

- The Python quasimodel reasoner (`src/alcircc5_reasoner.py`) is retained only as a cross-validation oracle. It is known-incomplete on cyclic-via-symmetric-role concepts (PO-loop, DR-loop, PP-PPI cycle, PPI-PP cycle): SAT answers are trustworthy, UNSAT answers are sound only for tree-model concepts.

- No complexity claim above doubly-exponential $B(C_0) = 2^{2^{p(n)}}$ is asserted for the full logic. The coarse manuscript bound is preserved. No EXPTIME upper bound is claimed.

Bottom line

The Python WP1–WP6 verification campaign reports **PASS** on every script; no counterexample to the repaired split-forest decidability proof was found. The 24-test WP2 corpus and the arity-4 WP6 tests are reproducible from commit `1f9b3b4` at `https://github.com/lambdamikel/alcircc5`. The multiple-PP-witness obligation (C5) is closed by the explicit C_{AB}^{inc} formula and the incomparability lemma in §2.2; the request-closure obligation (C6) is closed by the pure-isolation test in §2.3.

Three obligations remain open and are deferred to the technical addendum: the support-closed mosaic patchwork theorem (C7; addendum item 4), the standalone typed-EQ quotient lemma (C4; addendum item 5), and the finite certificate grammar with derived enumeration bound (C8; addendum item 6). Beyond the addendum, full publication-grade closure requires the Lean L1–L7 formalisation; the priority order on that stack remains L1, L3, L4, L6, then L2/L5/L7.

The revised verdict, following GPT-5.5's third-round review (lines 80–85), is: *promising and substantially strengthened verification evidence; no reported bounded counterexample; several proof-critical obligations remain open until the typed-equality quotient, request-closed lasso, support-closed mosaic patchwork, and finite certificate enumeration lemmas are made explicit.*